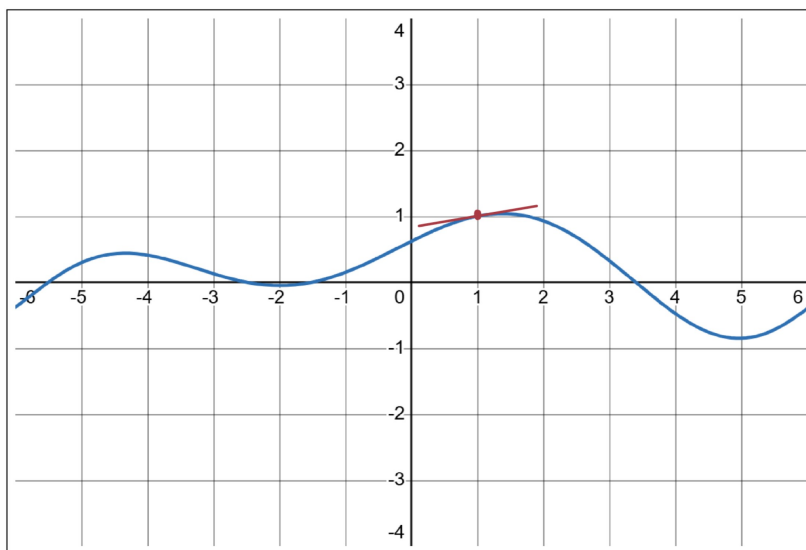


A4 final practice

1. (Fall 2024, #4) Consider the function f whose graph is shown below.



- (a) One of the following is the exact value of $f'(1)$. Pick which it is. Note: The peak of the graph occurs a little to the right of $x = 1$, not at $x = 1$.
- i. 100
 - ii. $\frac{1}{5}$
 - iii. $-\frac{1}{5}$
 - iv. -300
- (b) Use only your answer to part (a) and the fact that $f(1) = 1$ and no other information to estimate $f(0.9)$.

$$f(0.9) \approx f(1) + f'(1)(0.9 - 1) = 1 + \frac{1}{5} \cdot (-0.1) = \boxed{0.98}$$

- (c) Determine whether your approximation in (b) is an over or underestimate of the actual value of $f(0.9)$ and justify your answer. (Hint: use the given graph).

overestimate

because the graph of $f(x)$ lies below the linear approximation (tangent line)

2. (Spring 2024, #4) Consider the equation $\cos(xy) = x^2 + y$ and the curve it defines.

(a) Find an equation for the tangent line to the curve at the point $(1, 0)$.

$$\frac{d}{dx} [\cos(xy) = x^2 + y]$$

$$-\sin(xy)(y + xy') = 2x + y'$$

at $(x, y) = (1, 0)$:

$$0 = 2 + y' \Rightarrow y' = -2$$

so the tangent line at $(1, 0)$ is $y - 0 = -2(x - 1)$ (point-slope formula)

$$\boxed{y = -2(x - 1)}$$

(b) Use your answer (and the idea of linearization) to find an approximation for a y so that $(1.1, y)$ is on the curve.

linear approximation at $x = 1$:

$$L(x) = \underbrace{f'(1)}_{\substack{-2 \\ \text{from part (a)}}} (x - 1) + \underbrace{f(1)}_{y=0} = -2(x - 1)$$

$$L(1.1) = -2(1.1 - 1) = \boxed{-0.2}$$

3. (Fall 2023, #4) Given the relation

$$\frac{d}{dx} [e^{xy} + y \sin x + y^2 - 9 = 0]$$

(a) Find $\frac{dy}{dx}$ at $(0, 3)$.

$$e^{xy}(y + xy') + y' \sin x + y \cos x + 2yy' = 0$$

$$\text{at } (x, y) = (0, 3): 1 \cdot (3 + 0) + 0 + 3 + 6y' = 0 \Rightarrow y' = -1 \text{ at } (0, 3)$$

(b) Find the equation of the tangent to its graph at $(0, 3)$.

point slope formula: $y - 3 = (-1) \cdot (x - 0)$

$$y = -x + 3$$

4. (Spring 2023, #3) Let $f(x) = (2x+2)^{\frac{2}{3}}$.

(a) Find the linearization of $f(x)$ at $x = 3$.

$$f'(x) = \frac{2}{3} (2x+2)^{-\frac{1}{3}} \cdot 2 = \frac{4}{3} (2x+2)^{-\frac{1}{3}}$$

$$f'(3) = \frac{4}{3} \cdot 8^{-\frac{1}{3}} = \frac{4}{3} \cdot \frac{1}{2} = \frac{2}{3}$$

$$f(3) = 8^{\frac{2}{3}} = 4$$

the linearization at $x=3$ is: $L(x) = f'(3)(x-3) + f(3)$
 $= \boxed{\frac{2}{3}(x-3) + 4}$

(b) Use your linearization from part (a) to approximate $(8.04)^{\frac{2}{3}}$.

$$2x+2 = 8.04 \Rightarrow x = 3.02$$

$$L(3.02) = \frac{2}{3}(3.02-3) + 4 = \boxed{\frac{0.04}{3} + 4}$$

(c) Determine the sign of the concavity of $f(x)$ at $x = 3$.

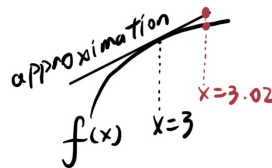
from (a), $f'(x) = \frac{4}{3} (2x+2)^{-\frac{1}{3}}$

$$f''(x) = \frac{4}{3} \left(-\frac{1}{3}\right) (2x+2)^{-\frac{4}{3}} \cdot 2$$

$$f''(3) = -\frac{8}{9} \cdot 6^{-\frac{4}{3}} < 0, \text{ the function is concave down at } x=3$$

(d) Use your answer from part (c) to determine if your estimate in part (b) is an overestimate or underestimate. You won't receive any credit unless you justify your answer using part (c).

Since $f''(3) < 0$, $f(x)$ is concave down at $x=3$, so the linear approximation line $L(x)$ lies above the graph of $f(x)$.



This causes the approximation to overestimate the actual value of $(8.04)^{\frac{2}{3}}$.

5. (Spring 2022, #5) Consider the curve $\frac{d}{dx}[e^{xy} + 3x^2 + y^2 = 65.]$

(a) Find the equation of the tangent line at the point $(0, 8)$.

$$e^{xy}(y + xy') + 6x + 2yy' = 0$$

$$\text{at } (x, y) = (0, 8): \quad 1 \cdot (8 + 0) + 0 + 16y' = 0 \Rightarrow y' = -\frac{1}{2}$$

so the tangent line at $(0, 8)$:

$$y - 8 = -\frac{1}{2}(x - 0)$$

$$\boxed{y = -\frac{1}{2}x + 8}$$

(b) There are two points on the curve with x -coordinate 0.1. Use part (a) to estimate one possible y -coordinate of the point $(0.1, y)$.

From (a), linear approximation at $x=0$ is: $L(x) = -\frac{1}{2}x + 8$.

$$L(0.1) = -\frac{1}{2} \cdot 0.1 + 8 = \boxed{7.95}$$